

CSc 8830 Computer Vision, Module 4, Question 3

Edge Detection and Region Segmentation in the Fourier Domain

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1 Setup: the 2D Fourier transform

For a continuous image $f(x, y)$ the Fourier transform and its inverse are

$$F(u, v) = \iint_{-\infty}^{\infty} f(x, y) e^{-j2\pi(ux+vy)} dx dy, \quad (1)$$

$$f(x, y) = \iint_{-\infty}^{\infty} F(u, v) e^{j2\pi(ux+vy)} du dv, \quad (2)$$

with (u, v) in cycles per unit length. For an $M \times N$ digital image, the discrete Fourier transform (DFT) is

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(\frac{ux}{M} + \frac{vy}{N})}, \quad u = 0, \dots, M-1, v = 0, \dots, N-1. \quad (3)$$

We write $\rho = \sqrt{u^2 + v^2}$ for radial frequency.

Convolution theorem. Every result below rests on this identity. For $g = h * f$,

$$G(u, v) = \iiint h(s, t) f(x - s, y - t) e^{-j2\pi(ux+vy)} ds dt dx dy.$$

Substituting $x' = x - s$, $y' = y - t$ splits the exponential into $e^{-j2\pi(us+vt)} e^{-j2\pi(ux'+vy')}$, and the integral factors:

$$\boxed{g = h * f \iff G(u, v) = H(u, v) F(u, v)} \quad (4)$$

Any linear, shift-invariant filter therefore acts in the frequency domain as a pointwise multiplication by its transfer function $H(u, v)$. Designing an edge detector reduces to choosing H .

2 Why edges live at high frequencies

2.1 The derivative theorem

Differentiate the inverse transform (2) with respect to x . Only the exponential depends on x :

$$\frac{\partial f}{\partial x} = \iint F(u, v) \frac{\partial}{\partial x} e^{j2\pi(ux+vy)} du dv = \iint [j2\pi u F(u, v)] e^{j2\pi(ux+vy)} du dv.$$

The bracket is, by definition, the transform of $\partial f / \partial x$. Repeating for y :

$$\mathcal{F}\left\{\frac{\partial f}{\partial x}\right\} = j2\pi u F(u, v), \quad \mathcal{F}\left\{\frac{\partial f}{\partial y}\right\} = j2\pi v F(u, v). \quad (5)$$

The gradient filters $H_x = j2\pi u$ and $H_y = j2\pi v$ have magnitude growing linearly with frequency: they suppress slowly varying content and amplify rapid change. Differentiation is high-pass filtering.

2.2 The spectrum of an edge

Model a vertical step edge of contrast A at x_0 as $f(x) = A \mathcal{H}(x - x_0)$, where $\mathcal{H}$ is the unit step. Using $\mathcal{F}\{\mathcal{H}(x)\} = \frac{1}{j2\pi u} + \frac{1}{2}\delta(u)$ and the shift theorem,

$$F(u) = A e^{-j2\pi u x_0} \left[\frac{1}{j2\pi u} + \frac{1}{2}\delta(u) \right], \quad |F(u)| = \frac{A}{2\pi|u|} \text{ for } u \neq 0. \quad (6)$$

A step spreads energy over all frequencies with a slow $1/|u|$ decay, far slower than the rapid falloff of a smooth region. Applying (5):

$$\mathcal{F}\{f'(x)\} = j2\pi u \cdot \frac{A e^{-j2\pi u x_0}}{j2\pi u} = A e^{-j2\pi u x_0} \iff f'(x) = A \delta(x - x_0).$$

The derivative flattens the spectrum to constant magnitude A , and the inverse transform concentrates all of it at x_0 : a spike at the edge whose height encodes the contrast. The linear phase $e^{-j2\pi u x_0}$ carries the edge location.

2.3 Finite differences are approximations of $j2\pi u$

The central difference $d(x) = \frac{1}{2}[f(x+1) - f(x-1)]$ has transfer function

$$H_{\text{cd}}(u) = \frac{1}{2}(e^{j2\pi u} - e^{-j2\pi u}) = j \sin(2\pi u) \approx j2\pi u \text{ for small } u.$$

The Sobel operator multiplies this by a $[1 \ 2 \ 1]/4$ smoother along the other axis, whose transform is $\frac{1}{4}(2 + 2 \cos 2\pi v) = \cos^2(\pi v)$, so

$$H_{\text{Sobel},x}(u, v) \propto j \sin(2\pi u) \cos^2(\pi v). \quad (7)$$

Sobel is a derivative along u combined with a low-pass along v . The frequency view explains its noise resistance directly.

2.4 The Laplacian

Applying (5) twice in each direction and summing:

$$\mathcal{F}\{\nabla^2 f\} = [(j2\pi u)^2 + (j2\pi v)^2] F = -4\pi^2(u^2 + v^2) F = -4\pi^2 \rho^2 F. \quad (8)$$

The Laplacian is isotropic (it depends on ρ alone) and grows quadratically, so it is an even stronger high-pass. The discrete 5-point stencil gives $H = 2 \cos 2\pi u + 2 \cos 2\pi v - 4 = -4(\sin^2 \pi u + \sin^2 \pi v)$, which matches $-4\pi^2 \rho^2$ near the origin.

3 Noise, and why edge detectors need a band-pass

Write the observed image as $f = s + n$ with white noise n of power spectral density σ_n^2 (flat in frequency). After a filter H , the output noise power is

$$P_n = \sigma_n^2 \iint |H(u, v)|^2 du dv. \quad (9)$$

For the gradient, $|H|^2 = 4\pi^2\rho^2$. In polar coordinates up to a band limit $\rho_{\max}$,

$$P_n = \sigma_n^2 \int_0^{2\pi} \int_0^{\rho_{\max}} 4\pi^2\rho^2 \rho d\rho d\theta = 2\pi^3\sigma_n^2\rho_{\max}^4.$$

Noise power grows as $\rho_{\max}^4$ (and as $\rho_{\max}^6$ for the Laplacian) while the edge signal in (6) decays as $1/\rho$. At high frequencies noise wins, so a practical detector must cap the gain at high ρ . The standard choice is a Gaussian low-pass:

$$g_\sigma(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \iff G_\sigma(u, v) = e^{-2\pi^2\sigma^2(u^2+v^2)}. \quad (10)$$

A Gaussian transforms to a Gaussian, with widths inversely related (σ in space, $1/(2\pi\sigma)$ in frequency).

3.1 Laplacian of Gaussian (LoG) as a band-pass filter

Smoothing then differentiating equals differentiating the kernel, $\nabla^2(g_\sigma * f) = (\nabla^2 g_\sigma) * f$. Combining (8) and (10):

$$H_{\text{LoG}}(\rho) = -4\pi^2\rho^2 e^{-2\pi^2\sigma^2\rho^2}. \quad (11)$$

This is zero at $\rho = 0$ (it ignores uniform brightness) and decays to zero at high ρ (it rejects noise). The peak response comes from

$$\frac{d}{d\rho} (\rho^2 e^{-a\rho^2}) = 2\rho e^{-a\rho^2} (1 - a\rho^2) = 0, \quad a = 2\pi^2\sigma^2 \implies \boxed{\rho^* = \frac{1}{\sqrt{2}\pi\sigma}}$$

so σ selects the scale of edge the detector responds to most. For $\sigma = 2$ px, $\rho^* \approx 0.113$ cycles/px (Figure 2). The difference of two Gaussians, $G_{\sigma_1} - G_{\sigma_2}$, approximates the same band-pass shape and is how SIFT builds its scale space.

3.2 Edges are the zero crossings

In 1D, a smoothed step is $A\Phi((x - x_0)/\sigma)$ with Φ the Gaussian CDF. Its second derivative is

$$\frac{d^2}{dx^2} A\Phi\left(\frac{x-x_0}{\sigma}\right) = -\frac{A(x-x_0)}{\sqrt{2\pi}\sigma^3} e^{-\frac{(x-x_0)^2}{2\sigma^2}},$$

which changes sign at $x = x_0$, where the first derivative peaks. The LoG output therefore crosses zero exactly on the edge, and the slope of the crossing scales with the contrast A . Thresholding that slope rejects weak crossings from noise.

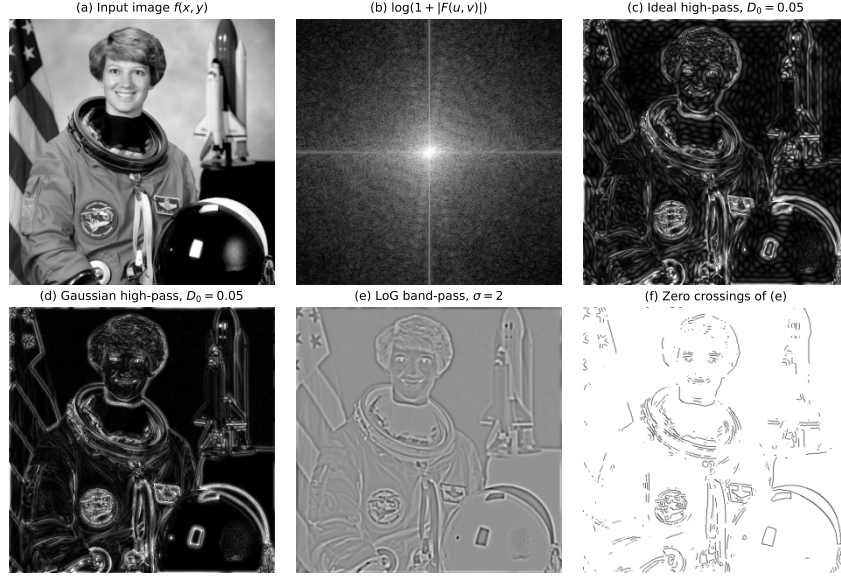


Figure 1: Frequency-domain edge detection on a 512×512 image, every filter applied as $G = HF$ followed by an inverse FFT. (b) Most energy sits near the origin; the bright cross comes from the image borders under the DFT’s periodic assumption. (c,d) High-pass filters keep edges and discard smooth regions. (e,f) The LoG band-pass of Eq. (11) and its zero crossings trace edge contours.

4 Frequency-domain filter design and the ringing problem

A high-pass filter is one minus a low-pass: $H_{\text{HP}} = 1 - H_{\text{LP}}$, so $h_{\text{HP}} = \delta - h_{\text{LP}}$. With cutoff D_0 and $D = \rho$, three common choices are:

$$\begin{aligned} \text{Ideal: } H &= \begin{cases} 0, & D \leq D_0 \\ 1, & D > D_0 \end{cases} \\ \text{Butterworth (order } n\text{): } H &= \frac{1}{1 + (D_0/D)^{2n}} \\ \text{Gaussian: } H &= 1 - e^{-D^2/(2D_0^2)} \end{aligned}$$

Why the ideal filter rings. The ideal low-pass is a disc of radius D_0 in frequency. Its inverse transform, computed in polar coordinates with the Bessel identity $\int_0^{2\pi} e^{j2\pi\rho r \cos\theta} d\theta = 2\pi J_0(2\pi\rho r)$, is

$$h_{\text{LP}}(r) = \int_0^{D_0} 2\pi\rho J_0(2\pi\rho r) d\rho = \frac{D_0 J_1(2\pi D_0 r)}{r}, \quad (12)$$

a “jinc” with oscillating side lobes. Convolution with it reproduces those oscillations as parallel ghost edges (the 2D Gibbs phenomenon). The Gaussian low-pass inverts to $2\pi D_0^2 e^{-2\pi^2 D_0^2 r^2}$, which is positive everywhere with no side lobes, so the Gaussian high-pass produces no ringing. Figure 2 shows the difference on a real image.

4.1 Algorithm

1. Pad the $M \times N$ image to $P \times Q = 2M \times 2N$. The DFT treats the image as periodic, and padding prevents wraparound, where the right edge convolves with the left.
2. Multiply by $(-1)^{x+y}$ to centre the spectrum, then compute $F(u, v)$ with the FFT in $O(PQ \log PQ)$.
3. Multiply by the chosen $H(u, v)$: $j2\pi u$ and $j2\pi v$ times G_σ for a gradient, or H_{LoG} .
4. Inverse FFT, undo the $(-1)^{x+y}$, crop to $M \times N$.
5. Gradient route: threshold $|\nabla f|$, optionally with non-maximum suppression and hysteresis (Canny).
LoG route: keep zero crossings whose slope exceeds a threshold.

For large kernels, the FFT route costs $O(\log PQ)$ per pixel against $O(k^2)$ for direct convolution with a $k \times k$ kernel.

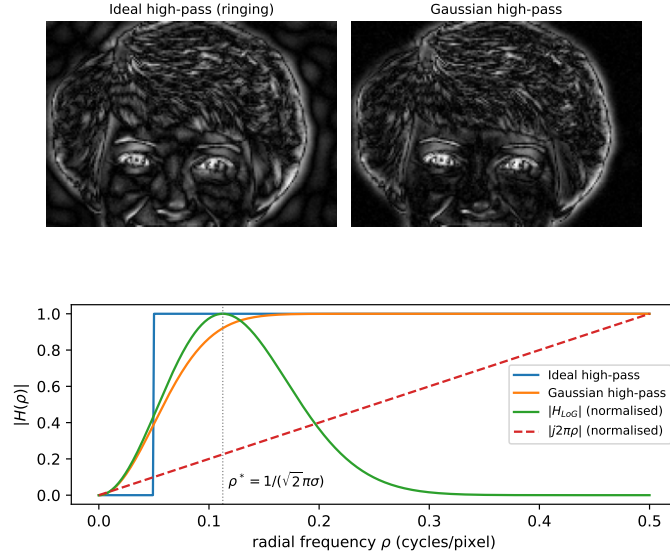


Figure 2: Top: zoom on the face. The ideal high-pass adds ripple bands parallel to each edge (the jinc side lobes); the Gaussian high-pass does not. Bottom: radial profiles $|H(\rho)|$. The pure derivative $|j2\pi\rho|$ keeps rising, so it amplifies noise; the LoG peaks at $\rho^* = 1/(\sqrt{2}\pi\sigma)$ and returns to zero.

5 Region segmentation in the frequency domain

Edges and regions are two views of one spectrum. Inside a homogeneous region, intensity varies slowly and the local spectrum concentrates at low frequencies; at a boundary, the spectrum spreads out as in (6). Segmentation can therefore work from either side.

5.1 Boundary route: close the edge map

The LoG zero crossings form closed contours by construction, because the zero set of a smooth function separates positive from negative regions. Labelling the connected components of $\{g_{\text{LoG}} >$

$0\}$ and $\{g_{\text{LoG}} < 0\}$ gives a region segmentation directly.

5.2 Region route 1: low-pass before thresholding

Intensity thresholding (Otsu, as used in Questions 1 and 2) needs two well-separated histogram modes. Noise widens each mode by σ_n . A Gaussian low-pass shrinks the noise variance to

$$\sigma_n^2 \iint e^{-4\pi^2 \sigma^2 \rho^2} du dv = \frac{\sigma_n^2}{4\pi\sigma^2},$$

so the modes narrow by a factor $2\sqrt{\pi}\sigma$ and the threshold separates regions more reliably. The price is a boundary blurred over roughly σ pixels.

5.3 Region route 2: texture segmentation with local spectra

When two regions share the same mean brightness but differ in texture, the global spectrum mixes them. We need a spectrum localised around each pixel, the windowed (short-time) Fourier transform:

$$F_w(x, y; u, v) = \iint f(s, t) w(s - x, t - y) e^{-j2\pi(us+vt)} ds dt. \quad (13)$$

With a Gaussian window, fixing one frequency (u_0, v_0) turns this into a convolution with a *Gabor filter*:

$$h_{u_0, v_0}(x, y) = e^{-\frac{x^2+y^2}{2\sigma^2}} e^{j2\pi(u_0x+v_0y)}. \quad (14)$$

Transfer function of the Gabor filter. By the modulation theorem, multiplying by $e^{j2\pi(u_0x+v_0y)}$ shifts the spectrum by (u_0, v_0) . Directly:

$$H(u, v) = \iint e^{-\frac{x^2+y^2}{2\sigma^2}} e^{-j2\pi[(u-u_0)x+(v-v_0)y]} dx dy = 2\pi\sigma^2 e^{-2\pi^2\sigma^2[(u-u_0)^2+(v-v_0)^2]}.$$

The Gabor filter is a Gaussian band-pass centred on (u_0, v_0) : it selects one spatial frequency and one orientation, $\theta_0 = \arctan(v_0/u_0)$.

Local energy features. For a bank of K filters, define

$$E_k(x, y) = (|h_k * f|^2) * g_{\sigma_E}(x, y), \quad k = 1, \dots, K. \quad (15)$$

By Parseval's theorem, $|h_k * f|^2$ measures the energy the neighbourhood of (x, y) carries in band k , and the extra Gaussian smoothing averages out the ripple of the carrier. Each pixel gets a feature vector $(E_1, \dots, E_K)$. A texture with a dominant frequency near (u_k, v_k) produces a large E_k , so pixels are assigned to the region whose band dominates. For two textures A and B the rule is

$$\text{pixel} \in B \iff \log E_B(x, y) - \log E_A(x, y) > 0, \quad (16)$$

and the boundary is the zero level set of this log-ratio. The log makes the rule invariant to local contrast, because a contrast scale multiplies both energies by the same factor.

Resolution trade-off. The window cannot be small in both domains. Measure spread as the standard deviation of $|h|^2$ in space and of $|H|^2$ in frequency. For the Gaussian envelope, $|h|^2 \propto e^{-x^2/\sigma^2}$ gives $\Delta x = \sigma/\sqrt{2}$, and $|H|^2 \propto e^{-4\pi^2\sigma^2 u^2}$ gives $\Delta u = 1/(2\sqrt{2}\pi\sigma)$, so

$$\Delta x \Delta u = \frac{1}{4\pi}, \quad (17)$$

the lower bound of the uncertainty principle. Gabor filters reach it exactly, which is why they are the standard choice. A large σ separates textures with close frequencies but localises the boundary only to about σ pixels; a small σ locates the boundary sharply but confuses similar textures.

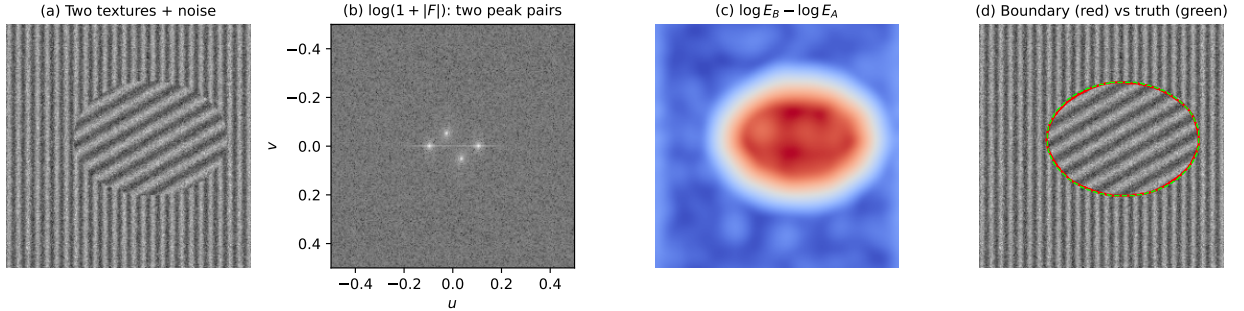


Figure 3: Texture segmentation with Gabor energy. (a) An elliptical region of oblique stripes (0.06 cycles/px, 60°) inside vertical stripes (0.10 cycles/px), plus Gaussian noise with standard deviation 0.6 against a stripe amplitude of 1. (b) Each texture appears as a symmetric pair of spectral peaks. (c) Two Gabor band-passes, each built in the frequency domain as a pair of Gaussians at $\pm(u_0, v_0)$ with $\sigma = 6$, give energies whose log-ratio separates the regions. (d) The zero level set (red) matches the true boundary (dashed green); 99.6% of pixels are labelled correctly.

5.4 Phase congruency: edges from the phase spectrum

Magnitude-based detectors depend on contrast. The phase spectrum offers an alternative. Expand a 1D signal locally as $f(x) = \sum_n A_n \cos(n\omega x + \phi_n)$ and define

$$PC(x) = \frac{|\sum_n A_n e^{j(n\omega x + \phi_n)}|}{\sum_n A_n} \in [0, 1]. \quad (18)$$

By the triangle inequality $PC = 1$ exactly when every component has the same local phase. A step has the Fourier series $\frac{4}{\pi} \sum_k \frac{\sin((2k+1)\omega x)}{2k+1}$, and at $x = 0$ every term sits at the same phase (a rising zero crossing), so $PC(0) = 1$ at the edge. Scaling the contrast multiplies every A_n by the same factor and leaves PC unchanged, so phase congruency finds low-contrast boundaries that a gradient threshold misses.

6 Summary

Operation	Transfer function $H(u, v)$	Role
Gradient	$j2\pi u, j2\pi v$	high-pass, edge strength and direction
Laplacian	$-4\pi^2 \rho^2$	isotropic high-pass, zero crossings at edges
Gaussian smoothing	$e^{-2\pi^2 \sigma^2 \rho^2}$	low-pass, noise suppression
LoG	$-4\pi^2 \rho^2 e^{-2\pi^2 \sigma^2 \rho^2}$	band-pass, peak at $\rho^* = 1/(\sqrt{2}\pi\sigma)$
Gaussian high-pass	$1 - e^{-\rho^2/(2D_0^2)}$	edge enhancement without ringing
Gabor	$2\pi\sigma^2 e^{-2\pi^2 \sigma^2 [(u-u_0)^2 + (v-v_0)^2]}$	frequency and orientation band-pass, texture

Edges carry the high-frequency energy of an image, because the derivative theorem multiplies the spectrum by $j2\pi u$ and a step's spectrum decays only as $1/\rho$. Noise forces the detector to be a band-pass instead of a pure high-pass, and the LoG's parameter σ picks the edge scale. For segmentation, low-pass filtering sharpens intensity histograms for thresholding, while local band energies from Gabor filters separate regions that differ in texture, with the uncertainty principle setting the trade-off between spectral discrimination and boundary accuracy.